IoT-Based Bus Stop System for Public Transport Overcrowding Prediction Using Machine Learning

CHAPTER 1
INTRODUCTION AND COMMUNITY CONTEXT

1.1 Project Background

This project is intended for daily urban commuters, particularly students, office workers, elderly passengers, and transit staff who depend on municipal bus services for routine travel. Their schedules, safety, and productivity are closely connected to the reliability of public transport. The project therefore addresses a broad and active community whose members regularly experience the conditions at crowded bus stops.

The project belongs to the Smart City and Internet of Things (IoT) domain, supported by Artificial Intelligence (AI) for predictive analysis. It applies low-cost sensing, edge processing, cloud communication, and machine learning to a public-transport setting. The main technical focus is not the bus itself, but the physical waiting point where passengers gather before boarding.

This area was selected because transport modernization has largely concentrated on vehicle tracking, electronic ticketing, and route information, while many bus shelters remain passive structures. A standard stop normally provides no automated measurement of passenger build-up and no way to communicate local crowding conditions. Converting such a stop into a connected sensing node creates an opportunity to improve awareness without major construction or expensive industrial equipment.

The project is modeled around Dindoshi Bus Depot and the BEST Route 326 boarding area in Malad East, Mumbai. This is a dense urban setting with fixed transit corridors, monsoon exposure, and strong morning and evening demand peaks. These characteristics make it a suitable location for developing and evaluating an affordable prototype that can later be adapted to similar municipal bus stops.

1.2 Community Problem Identification

A recurring difficulty at busy urban bus stops is severe and unpredictable overcrowding within a limited shelter area. Passenger numbers can rise quickly during office and college rush hours, while heavy rain can bring additional people under the shelter. When the waiting area reaches capacity, physical comfort declines and orderly boarding becomes difficult.

The issue was first identified through direct personal experience as a regular commuter and through repeated observation of passenger movement at Dindoshi Bus Depot, especially near the BEST Route 326 boarding edge. The problem was most visible during the morning peak from 8:00 to 10:00 AM, the evening peak from 5:00 to 7:00 PM, and periods of heavy monsoon rain. Informal conversations with commuters and depot staff confirmed that the situation was familiar rather than exceptional.

Daily office workers and students are affected most frequently because they travel during the same concentrated time windows. Elderly passengers and people carrying bags are particularly vulnerable when the shelter and pavement become congested. Depot staff are also affected because they must judge crowd severity visually and respond only after conditions have already become difficult to manage.

The visible signs include passengers standing beyond the designated shelter, obstruction of the adjoining pavement, movement close to active traffic, heat and discomfort inside the shelter, and a sudden scramble when a bus arrives. These conditions can increase boarding time and contribute to delays along the route. The absence of an automated local count or warning means that neither approaching commuters nor transit staff receive early notice of the developing crowd.

The identified issue is therefore not simply the presence of many passengers. It is the lack of timely, objective information about how quickly a localized crowd is building and whether the current condition is normal, cautionary, or critical. This observation provided the basis for the evidence-gathering activities described in the following section.

1.3 Community Need Assessment

To establish whether the observed crowding problem represented a genuine community need, the assessment combined direct observation, a Google Forms survey, and informal staff interviews. Direct observation documented crowd build-up, weather exposure, bus departures, and the limitations of relying on visual judgement, while the survey and interviews collected the practical expectations of commuters and depot staff.

The Google Forms survey was completed by exactly 25 regular commuters at the Dindoshi Bus Depot / BEST Route 326 boarding area, and three depot staff members participated in informal interviews. Commuters were asked whether they preferred an on-site indicator or a mobile-application-only approach, how quickly a critical warning should appear, and whether the passenger count should be reduced or completely reset after a bus departure. Staff discussions focused on remote monitoring and controlled count correction.

The recorded responses provided direct quantitative evidence of the need. Of the 25 commuters, 21 (84%) preferred an on-site indicator rather than relying only on a mobile application, 23 (92%) wanted a critical warning within 30 seconds, and 19 (76%) wanted the passenger count reduced after a bus departure rather than reset to zero. The staff interviews also identified the need for remote crowd alerts and an operator-controlled method of correcting the displayed count. These results show that the requirement was shared by a clear majority of respondents rather than being an isolated personal preference.

The assessment therefore revealed a clear gap between the passive bus stop facilities currently available and the information the community actually requires. Commuters lack early, visible crowd-status information, while depot staff lack continuous remote monitoring and a controlled correction mechanism after a bus leaves. The evidence supports the need for an affordable IoT-based system that can sense crowd build-up, communicate timely warnings, and assist staff without depending entirely on manual observation.

1.4 Existing System Analysis

At present, crowding at the selected bus stop is managed mainly through manual visual assessment. Commuters discover the condition only after reaching the stop, while depot staff estimate the crowd by observation or receive information from drivers when boarding becomes disorderly. This method requires a person to be physically present and provides no continuous record of how the crowd developed.

The principal limitation is that the response is reactive. By the time severe congestion is reported, passengers may already be standing outside the shelter and boarding delays may already have occurred. Visual estimates can also differ between observers and do not provide a consistent threshold for distinguishing normal, warning, and critical conditions. Without a data history, staff cannot easily review recurring patterns or evaluate how weather and time affect the stop.

Existing transit applications partially support commuters through timetables, route information, and GPS-based bus tracking. These tools can show when a bus may arrive, but they do not measure the waiting crowd at the shelter. Industrial automatic passenger-counting products exist, but they are generally installed inside vehicles, use cameras or commercial infrared systems, and are costly for a small stop-level prototype. Low-cost experimental counters often provide only a raw count without weather context, risk classification, controlled alerting, or operator correction.

Consequently, the existing approach does not provide anonymous stop-level sensing, predictive risk assessment, local public indication, remote authority alerts, and persistent records in one affordable system. A solution must improve awareness without requiring commuters to install an application or exposing identifiable personal data.

1.5 Problem Statement

Daily commuters at busy urban bus stops face unpredictable localized overcrowding during peak travel periods and adverse weather, while the current manual approach provides no timely or objective measurement of crowd severity. This creates unsafe waiting and boarding conditions and prevents transit staff from responding proactively. A low-cost system is therefore required to sense passenger build-up, evaluate risk, and communicate clear local and remote warnings before conditions become critical.

1.6 Proposed Solution

The proposed solution is an IoT-based bus stop crowding prediction system that measures confirmed passenger crossings, monitors temperature, obtains live weather information, and uses a Random Forest regressor to estimate crowding risk. An ESP32 DoIT DevKit V1 acts as the edge controller, while a local Python backend performs prediction, classification, alerting, and logging.

A VL53L0X confirms a crossing only after two consecutive readings below 400 mm, reducing single-frame noise. A DHT22 supplies temperature, and OpenWeatherMap supplies current weather. The backend combines hour, rain status, passenger count, and temperature, then classifies the result as NORMAL (0-40), WARNING (41-75), or CRITICAL (above 75).

Green and red LEDs provide on-site indication, Blynk presents live values to staff, and Telegram alerts only for CRITICAL conditions. V7 subtracts 30 passengers with a floor of zero, V8 shows minutes since the last crossing, and each five-second cycle is logged locally. The low-cost, camera-free design can be fitted to an existing stop and combines anonymous sensing, environmental context, operator control, and local and remote communication in one prototype.

1.7 Aim and Objectives

Aim: To develop a low-cost, context-aware smart bus stop system that uses IoT sensing and machine learning to predict and communicate public-transport overcrowding risk.

Objectives:

- Design an ESP32 sensing unit with VL53L0X and DHT22 modules and two-reading confirmation.
- Integrate OpenWeatherMap data and map rain conditions into the backend features.
- Develop a 100-tree RandomForestRegressor using 43,200 formula-generated samples.
- Implement Blynk, red/green LEDs, and CRITICAL Telegram alerts with a 60-second cooldown.
- Implement V7 correction, V8 crossing evidence, and five-second CSV logging.
- Test all thresholds, virtual-pin ownership, timing, and alert rules.

1.8 Scope of the Project

The project covers the engineering of one functional bus-stop prototype for the Dindoshi / BEST Route 326 context. It includes single-direction crossing detection, local temperature measurement, live weather retrieval, machine-learning risk scoring, three-state classification, physical LED indication, Blynk monitoring, Telegram escalation, operator correction, and local CSV logging.

The system serves two principal groups: commuters who benefit from a glanceable on-site indication and transit staff who require remote visibility and controlled intervention. Its boundary begins at the sensor-monitored crossing point, continues through ESP32 filtering and Blynk communication to the local brain.py backend, and ends at the LEDs, Blynk dashboard, Telegram alert, and bus_stop_log.csv record.

Development and demonstration are limited to a single representative stop. The Python backend runs on a local Windows computer, and Blynk Cloud is used as the data bridge rather than as a permanent database. The prototype demonstrates the complete workflow but does not claim city-wide deployment or automatic control of transit operations.

1.9 Limitations

The project does not include bidirectional entry/exit tracking, camera-based identification, physical redesign of the shelter, automated vehicle dispatch, ticketing integration, a cloud database, or multi-stop synchronization. A single VL53L0X cannot determine whether a detected person is entering or leaving and may count repeated movement through the beam; it therefore records confirmed crossings rather than verified occupancy.

The system depends on Wi-Fi for Blynk, OpenWeatherMap, and Telegram communication. During a network interruption, the ESP32 can continue raw sensing, but an updated weather-enriched ML score, dashboard state, Telegram alert, and fresh risk-driven LED indication are not guaranteed until communication is restored. The local Python computer must also remain powered for backend prediction and CSV logging.

The 45-person shelter capacity and the 30-person V7 boarding subtraction are design estimates rather than measured occupancy values. The Random Forest model is trained on formula-generated data, so it approximates the transparent design formula but has not yet been validated against long-term field observations. Sensor placement, occlusion, stationary objects, and unusual crowd movement may also affect counting accuracy in real use.

1.10 Major Modules

The proposed system is organized into five connected functional modules:

- **Sensing and Edge Processing Module:** The ESP32 reads the VL53L0X on GPIO21/22 and the DHT22 on GPIO4, filters crossings, and publishes passenger count (V1), temperature (V4), and minutes since last crossing (V8).
- **Backend Machine Learning Module:** brain.py runs every five seconds, reads V1 and V4, builds [hour, is_raining, passenger_count, temperature], predicts risk with the Random Forest model, and classifies the result.
- **Communication and Cloud Bridge Module:** Blynk exchanges virtual-pin data between the ESP32 and Python; OpenWeatherMap provides weather by HTTPS; Telegram delivers CRITICAL alerts. ESP32 owns V1, V4, and V8, while Python owns V2, V3, V5, and V6.
- **Alert and Indication Module:** The red LED on GPIO25 represents CRITICAL and the green LED on GPIO26 represents NORMAL. Both use 220-ohm resistors. Telegram has a 60-second cooldown, and the physically wired buzzer remains disabled.
- **Visualization and Logging Module:** The Blynk dashboard displays V1-V8, including the V7 Bus Left (-30) operator control, while every backend cycle appends timestamp, passengers, temperature, weather, risk, and status to bus_stop_log.csv.

Together, these modules form a continuous pipeline from anonymous physical sensing to contextual risk evaluation, local and remote communication, controlled operator action, and persistent evidence.

1.11 Success Criteria

The prototype will be considered successful when the following measurable conditions are satisfied:

- Two consecutive VL53L0X readings below 400 mm produce one confirmed crossing, while a single noisy frame produces no count.
- DHT22 readings update at the planned three-second interval, invalid values are rejected, and the backend completes its nominal five-second cycle.
- Risk classification remains NORMAL at 40 or below, WARNING from 41 through exactly 75, and CRITICAL only above 75.
- Blynk V1-V8 display the correct live values without ESP32/Python ownership conflicts, and V7 subtracts 30 with a floor of zero.
- CRITICAL activates the red LED and Telegram alert, NORMAL activates the green LED, and repeated Telegram messages are suppressed for 60 seconds.
- bus_stop_log.csv receives one complete six-field record during every successful processing cycle.

1.12 Expected Outcomes

The immediate expected outcome is improved situational awareness at the bus stop. Commuters can understand whether the condition is normal or critical through the on-site LEDs without installing an application, while transit staff can view passenger count, weather, temperature, risk, and status remotely through Blynk. Telegram provides an additional escalation path when the risk becomes critical.

The V7 and V8 controls are expected to make operator intervention more defensible by allowing staff to reduce the estimate after a departure while retaining evidence of recent crossing activity. CSV logging creates a traceable record of each processing cycle, helping the team demonstrate the system, identify recurring periods of crowd build-up, and investigate abnormal readings.

Over time, accumulated field data can support retraining and validation against actual observations instead of relying only on formula-generated samples. The modular architecture also provides a foundation for future stop-level expansion, cloud database integration, and more accurate bidirectional sensing, subject to further development and municipal approval.

1.13 Summary

This chapter established the Smart City and IoT context of the project and identified unpredictable bus-stop overcrowding as a recurring safety and operational problem at the Dindoshi / BEST Route 326 setting. Observation, a 25-commuter survey, and three staff interviews confirmed the need for local indication, rapid remote alerts, and controlled count correction. The current manual approach and partial transit tools were found to lack anonymous stop-level sensing and predictive risk information. The chapter then presented the ESP32, VL53L0X, DHT22, OpenWeatherMap, Random Forest, Blynk, LED, Telegram, V7/V8, and CSV-based solution, followed by its aim, objectives, scope, limitations, modules, success criteria, and expected outcomes. The reader is now prepared to examine the supporting literature, comparative analysis, and detailed requirements in Chapter 2.